Science: Conjectures and Refutations

Karl R. Popper - Summary by Mahdi Rasouli

Popper was a philosopher who wanted to know what it is that makes a theory "**scientific**." He was very concerned with how to tell the difference between actual science and unreal appearances of science, like astrology or some of the stuff from psychology. He knew that real science can make mistakes, but "*pseudo-science*" can be wrong and pose as being right. He considered examples such as Einstein's theory of relativity, which functioned effectively, and Freud's and Marx's theories, which he believed were otherwise.

Popper's fundamental idea was that a scientific theory **must be falsifiable**, i.e., you must be capable of conceiving of some situation where the theory can be proven false. If you cannot think of any situation where the theory can be proven false, then it is not a scientific theory. **Real science makes bold predictions.** If the predictions are incorrect, then the theory is wrong. Good scientific theories say that things can't happen. The more things it says can't happen, the better the theory.

Popper applied the case of Einstein's theory. The theory predicted how gravity would change light. Scientists could experiment with this prediction. If the experiment failed, the theory would be false. But theories like those of Marx, Freud, and Adler were not like this. They seemed to explain everything. But Popper believed that they could explain anything, even the opposite of a behavior, and thus could not be falsified. That was a fault, not an advantage.

Popper continued that it's easy to find things that seem to verify that a theory is true. But **real verification is in trying to show that the theory is false**. If the theory can withstand many serious tests, then we have real support. Scientific theories tend to begin life as "*myths*." Just because something is not scientific, it doesn't mean that it is of no value whatsoever. But it can't pretend that it's founded on real evidence.